

LumenTrace: A high-performance MCU-based LED racing simulator

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Simon Völkl, Erik Dombrowsky, Simon Frey 
Department of Electrical Engineering, Media, and Computer Science
Ostbayerische Technische Hochschule Amberg-Weiden
Amberg, Germany

Abstract—Background: tbd **Methods:** tbd

Conclusion: tbd

Index Terms—MCU, LED, neopixel, WS2812, racing simulator, Raspberry Pi

VI. EVALUATION

A. Performance

tbd

B. Latency

tbd

C. Challenges

tbd

VII. CONCLUSION AND FUTURE WORK

tbd

I. INTRODUCTION

A. Motivation

tbd

B. Unique Selling Point & Contributions

tbd

C. Outline

tbd

II. RELATED WORK

tbd

III. SYSTEM ARCHITECTURE & REQUIREMENTS

A. Functional and Non-Functional Requirements

tbd

B. High-Level Architecture

tbd

IV. HARDWARE IMPLEMENTATION

A. Microcontroller Setup

tbd

B. Communication

tbd

C. Signal Processing & Controller

tbd

V. SOFTWARE IMPLEMENTATION & GAME MECHANICS

A. Core Loop

tbd

B. Physics & Lane Handling

tbd

C. Dev/Simulation Environment

tbd